



Intention-driven adaptive impedance control of human-robot collaboration for open space tasks

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Abstract

In the field of human-robot collaboration, enhancing the performance of interaction tasks in open space remains a critical challenge. Human-robot interaction based on surface electromyography (sEMG) has been attracted wide attention driven by the potential motion intentions. To improve the performance of robots performing interaction tasks in open space, a method based on a neuromusculoskeletal model and joint variable impedance controller is proposed, which is driven by intentions inferred from sEMG signals. First, a human upper limb musculoskeletal model is developed from the sEMG signal and calibrated to match the optimal motor behavior of the operator. Then, a joint variable impedance control algorithm is designed. Finally, an experimental study is conducted to perform drag trajectory tracking motion in open space, and the proposed method is experimentally compared with the constant impedance method of robots. The results show that the operator's energy consumption is reduced by 6.28 J, the average force is reduced by 3.83 N, and the task execution time is reduced by 5.25s under the variable-impedance approach compared to high-impedance. The trajectory tracking error is also reduced by 1.4 mm for variable-impedance compared to low-impedance. Our method using intention-driven adaptive impedance control balances the robot's performance metrics between fast free motion and precise tracking motion in complex tracking motion tasks.

Keywords Impedance control · Human-robot interaction · Open space task · sEMG-based intention · Neuromusculoskeletal model

1 Introduction

Collaborative operation between humans and robots is one of the directions for next-generation robotic systems, aiming to reduce the burden on the operator and improve efficiency. Achieving a robot's understanding of human operational intent is a key issue in human-robot interaction.

Various control methods have been proposed to improve the performance of human-robot interaction. For example, impedance control is applied to flexible articulated robots that interact with unknown environments (Ott et al. 2008). Or it is applied to a mechanical leg running at high speed to improve its regulation (Guo 2017). Rehabilitation robotics with impedance control can mechanical leg running at high speed to improve its regulation facilitate patient recovery to some extent (Zheng and Shen 2015).

The conventional constant impedance gain control mentioned above, while capable of demonstrating a certain level of compliance, falls short in enhancing track accuracy

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during the execution of dynamic and complex tasks (Liu 2020; He et al. 2016).

To improve the performance of human-robot interaction, motion intention estimated by sensor signals, such as force signals and velocity signals can be applied into control schemes. The current tuning of impedance parameters based on single force information only partially reflects the operational intent (Park et al. 2021), which inevitably makes the human-robot interaction unnatural. There are also various signals such as angular velocity and linear velocity to adjust the impedance parameters for continuous control, but the impedance regulation equations are determined experimentally and cannot be applied to other experiments of the same type (Sanchez-Magos et al. 2019). The concept of standard forces is introduced to modulate the impedance parameters to enable direct human intent estimation under fixed task trajectories (Kang 2019). These control methods struggle to determine optimal impedance parameter planning due to modeling inaccuracies, and are not accurate in reflecting the human operation intention.

Compared to impedance control for fixed task trajectories, open-space free-trajectory tasks do not have a fixed task trajectory and cannot be pre-trained with impedance parameter change rules, so the robot performs poorly in such tasks. Humans can adjust the impedance of the arm to the appropriate trajectory of movement by changing the level of muscle activation (Park et al. 2019; Li et al. 2017). The human body impedance information extracted from the sEMG signal can most accurately reflect the human movement intention (Buscaglione et al. 2024; Ahn et al. 2023; Doornebosch et al. 2021), which can be used as a control signal to the robot controller for the transfer of human operation skills (Zhu et al. 2022). This presents a feasible solution for enabling robots to tackle intricate interactive tasks (Arnold and Lee 2021).

In addition to the above methods to model the relationship between sEMG signals and stiffness a black box approach can be used including reinforcement learning (Bogdanovic et al. 2020), neural networks (He and Dong 2003; Kong et al. 2019), etc. However, reinforcement learning and neural network-based approaches have the disadvantages of requiring a large number of samples for training, high training costs, and poor task generalization.

Research on sEMG-based variable impedance control has been carried out, for example, by calculating the impedance value of the arm from the sEMG signal for changing the control parameters of a robot (Grafakos et al. 2016) or rehabilitation device (Huo et al. 2019). The above methods are difficult to obtain satisfactory performance due to the inaccurate estimation of impedance values. This limitation can be adjusted by dynamic modeling of the neuromuscular and skeletal system. Most of the above tasks are performed

on fixed trajectories based on physical constraints. It is difficult to design for trajectory tasks in open space, while some performance metrics such as accuracy and compliance are rarely discussed.

In this study, a intention-driven adaptive impedance control scheme for joint space is proposed to enable the manipulator to modify between fast free motion and precise tracking motion. Similar to previous studies, the sEMG of the arm muscles is collected and the endpoint stiffness of the arm is calculated by the Hill musculoskeletal model (Rajagopal 2016), and then the Jacobian matrix is calculated from the real-time angles acquired by the posture sensor, thus converting the endpoint stiffness of the arm into joint stiffness. The stiffness matrix and the damping matrix constitute the target impedance model, and the variable impedance control of the robot joints is realized by the joint variable impedance control algorithm based on the predicted impedance value of sEMG, which is input to the robot controller together with the end external force. Different from previous dragging studies (Kang 2019; Wu 2019), joint impedance values are applied to dragging tasks with uncertain trajectories, and a variable impedance physical human-robot interaction interface is simultaneously developed to provide an intuitive and comfortable interaction. This intention-driven adaptive impedance of the joint space allows the operator to better react to the dragging desires during operation and thus precisely change the robot's impedance value. In this way, the operator can have better maneuverability when performing tasks in open space.

The paper is organized as follows. Section 2 explains the joint variable impedance control based on the sEMG signal. Section 3 presents the experimental design. The experimental results are described and discussed in Sect. 4, and the conclusions are given in Sect. 5.

2 Methodology

2.1 Model

To estimate the joint torque and stiffness generated during upper limb movements, a musculoskeletal model is built based on the sEMG signals collected from related muscles. Four muscles that dominate the movement of the shoulder and elbow joints are selected. The structure of the shoulder and elbow muscles and the locations of the electrodes are shown in Fig. 1. The computational flow of the model is depicted in Fig. 2, including the muscle activation dynamics, muscle contraction dynamics, and musculoskeletal geometry components. Specifically, muscle activation a_i is derived from the EMG processing of each muscle, and then muscle-fiber force F_i^{mt} is derived, resulting in the shoulder



Fig. 1 The location of the four muscles

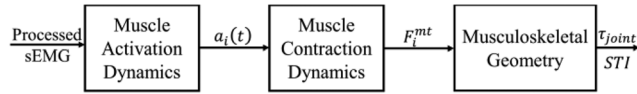


Fig. 2 Computational flow of the musculoskeletal model

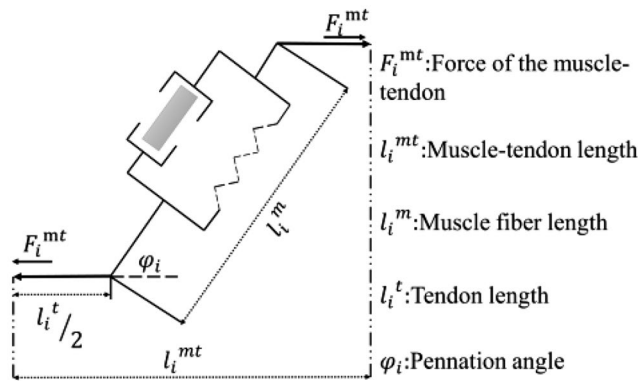


Fig. 3 Muscle-tendon unit model (Han et al. 2015)

stiffness trend STI . The details are to be discussed in the following.

In general, muscle activation dynamics can be expressed through the sEMG signals, and building an accurate musculoskeletal model requires the estimation of the sEMG muscle activation state. The muscle activation $a_i(t)$ of each muscle i is given by (Lloyd and Besier 2003; Trick 1975)

$$a_i(t) = \frac{e^{Au_i(t)} - 1}{e^A - 1} \quad (1)$$

where $u_i(t)$ is the neural activation of pre-processed EMG signal (Han et al. 2003; Trick 1975). A is the nonlinear shape factor varying between -3 and 0 .

After muscle activation is determined by the sEMG signal, muscle-tendon force changes need to be obtained based on muscle contraction characteristics, which are revealed by muscle contraction modeling. Most studies of muscle biomechanics are derived from the Hill muscle model, as shown in Fig. 3 and the muscle-tendon force $F_i^{mt}(t)$ can be calculated as (Han et al. 2003)

$$F_i^{mt}(t) = [f_A(l_i) f_V(v_i) a_i(t) + f_P(l_i)] \cos(\phi_i(t)) \quad (2)$$

where $f_A(l_i)$ is normalized force-length function, $f_V(v_i)$ is normalized force-velocity function, and $f_P(l_i)$ is normalized passive force-length function, $\phi_i(t)$ is the pennation angle. l_i is the normalized muscle fiber length, v_i is the normalized muscle fiber velocity, and

$$l_i = \frac{l_i^m}{l_0^m}, \quad v_i = \frac{v_i^m}{v_0^m} \quad (3)$$

where l_i^m and v_i^m are the muscle fiber length and the muscle contraction velocity, respectively; l_0^m represents the optimal fiber length, and v_0^m is the maximum muscle contraction velocity.

In order to calculate Eq. (2), a number of simplified methods have been proposed (Fleischer and Hommel 2008).

$$f_A(l_i) = \begin{cases} 1 - \left(\frac{l_i - 1}{0.5}\right)^2, & 0.5 \leq l_i \leq 1.5 \\ 0, & \text{otherwise} \end{cases} \quad (4a)$$

$$f_P(l_i) = e^{10l_i - 15} \quad (4b)$$

$$f_V(v_i) = 1 \quad (4c)$$

$\phi_i(t)$ is the pennation angle which is the angle between muscle fibers and tendons, and can be given by

$$\phi_i(t) = \sin^{-1} \left(\frac{l_{oi}^m \sin(\phi_{oi})}{l_i^m(t)} \right) \quad (5)$$

where, $l_i^m(t)$ is the muscle fiber length at time t , and ϕ_{oi} is the pennation angle at optimal muscle fiber length l_{oi}^m .

The $l_{oi}^m(t)$ is the optimal muscle fiber length for the muscle at maximum isometric muscle fiber force, which was defined by (Buchanan et al. 2004).

$$l_{oi}^m(t) = l_{oi}^m (\gamma (1 - a_i(t)) + 1) \quad (6)$$

where γ is the variation parameter, usually chosen as 0.15 (Visser 1990).

The muscle-tendon length l_i^{mt} is the sum of tendon length l_i^t and muscle fiber length l_i^m .

$$l_i^{mt}(t) = l_i^t(t) + l_i^m(t) \cos(\phi_i(t)) \quad (7)$$

With the tendon length determined, obtaining the muscle fiber length requires specifying the muscle-tendon unit length. The solution of the muscle-tendon unit length involves complex physiological parameters, which are simplified to a polynomial function of the joint angle (Visser 1990).

$$l_i^{mt}(t) = b_0 + b_1 \theta_i \quad (8)$$

where θ_i refers to the corresponding joint angle, b_0 and b_1 are constants.

Once the force of the muscle is derived, multiplying it with the moment arm of a musculotendinous unit gives the moment generated by that muscle at the joint. Using the displacement method (An et al. 2003), the equation for calculating the moment arms of a musculotendinous unit $r_{ai}(t)$ is defined by

$$r_{ai}(t) = \frac{\partial l_i^{mt}(t)}{\partial \theta_i} \quad (9)$$

When the muscle forces and moment arms are solved for, then the corresponding joint moments they produce can be calculated:

$$\tau_i = F_i^{mt} r_{ai}(\theta) \quad (10)$$

After determining the joint corresponding to all muscle-tendon forces and moment arms, the respectively generated joint torque $\tau_j(\theta, t)$ can be found

$$\tau_j(\theta, t) = \left| \sum_{i=1}^n \tau_i(t) \right|^{ago} - \left| \sum_{i=1}^n \tau_i(t) \right|^{anta} \quad (11)$$

where $\tau_i(t) = F_i^{mt} r_{ai}(t)$, and n is the number of agonist and antagonist muscles acting on the joint. τ^{ago} and τ^{anta} represent the torque generated by the agonist muscle and the antagonist muscle respectively.

Joint stiffness is the resistance of the limb to rotation under external forces, and the value of it is related to the joint moment τ_j and the stiffness trend index (STI), expressed as the sum of the values of each muscle moment.

$$STI(t) = \left| \sum_{i=1}^n \tau_i(t) \right|^{ago} + \left| \sum_{i=1}^n \tau_i(t) \right|^{anta} \quad (12)$$

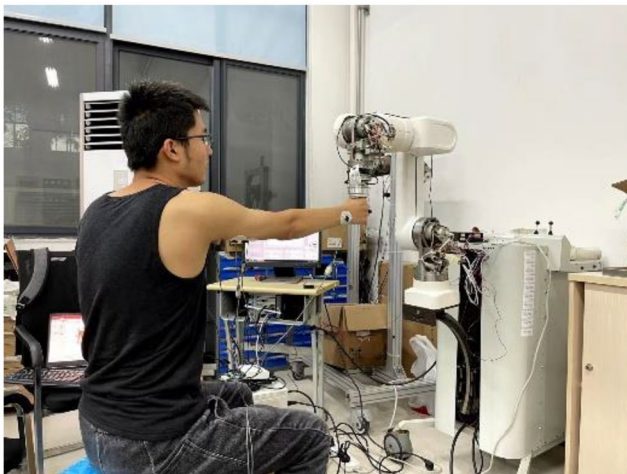


Fig. 4 Arm endpoint stiffness calibration experiment

The stiffness of the shoulder joint $K_l(t)$ is defined by

$$K_l(t) = \alpha_l \times STI(t) + \beta_l \quad (13)$$

where α_l and β_l are parameters to be determined, which will be obtained from the stiffness calibration experiments in the next section.

2.2 Model calibration and validation

2.2.1 Model calibration

According to Liang (2014), the relationship between the sEMG signal and joint torques $\tau(t)$ which is displayed by the six-dimensional force sensor, can be given by

$$\tau(t) = \sum_{i=1}^n \alpha_i A_i^{ago}(t) - \sum_{i=1}^n \beta_i A_i^{anta}(t) + r_\tau(t) \quad (14)$$

where α_i and β_i are the parameters found by least squares fitting. $A_i^{ago}(t)$ and $A_i^{anta}(t)$ are amplitude envelopes of the sEMG signal from the agonist muscle and the antagonist muscle, and $r_\tau(t)$ is the residual.

Muscle stiffness is approximately proportional to muscle torque, and the endpoint stiffness of the human arm can be the sum of the absolute values of joint muscle torque. The expression of the endpoint stiffness $K_P(t)$ can be given by

$$K_P(t) = \sum_{i=1}^n |\alpha_i| A_i^{ago}(t) + \sum_{i=1}^n |\beta_i| A_i^{anta}(t) + r_K(t) \quad (15)$$

where $r_K(t)$ denotes the nonlinear residual.

Once the endpoint stiffness of the arm is determined, the endpoint damping can be calculated from the stiffness. Damping $B_P(t)$ is related to the stiffness and the equation is given by

$$B_P(t) = 2\zeta \sqrt{K_P(t)} \quad (16)$$

where ζ is a constant that needs to be adjusted to suit the damping of the system.

At this point, the endpoint stiffness has been solved and converted to the joint stiffness K_q using the Jacobian matrix.

$$K_q(t) = \text{diag}[K_{q1}, K_{q2} \dots K_{qn}] \quad (17)$$

where $K_q(t) = J^T K_P(t) J$, J^T is the Jacobian matrix.

We design a stiffness calibration experiment to calibrate the endpoint stiffness. The operator's upper arm is perpendicular to the robotic manipulator's end in Fig. 4. The robot end is installed with a six-axis moment sensor (ATI-Axia80). The operator applied constant forces of 5 N, 9 N, and 13 N along six directions $[\pm X, \pm Y, \pm Z]$ with force display data

shown on the screen as visual feedback. The experiment is repeated 20 times in each direction, thus the parameters in Eq. (11) were estimated.

In the experiments, the sEMG signals of four muscles are recorded at a sampling frequency of 1024 Hz, followed by band-pass filtering from 20 to 500 Hz using a fourth-order Butterworth filter and trapped at 50 Hz to exclude industrial frequency interference. The operator performs a muscle maximum voluntary contraction (MVC) experiment first. Retaining the end six-dimensional force and four muscle sEMG signals, the parameters of Eq. (11) are solved by the least squares method. These parameters are used to calculate the endpoint stiffness in Eq. (12). Calculations following the above steps yielded the endpoint stiffnesses of the arm in the X, Y, and Z directions finally. Then, the conversion from endpoint stiffness to joint space stiffness is done using the Jacobian matrix.

2.2.2 Model validation

Model validation experiments are shown in Fig. 5, where the operator's shoulder joint is aligned with the torque sensor to perform flexion and extension movements of the shoulder joint. The stiffness of the torque sensor is set to 5 Nm/rad.

In the shoulder extension and flexion experiments, the torque estimated by the model and the torque curve measured by the torque sensor are shown in Fig. 6, respectively. To assess the accuracy of the model, we use the goodness-of-fit (*GoF*) (Ajoudani and Forero 2011) metric for evaluation.

$$GoF = 1 - \frac{\sqrt{\sum_{k=1}^N (\tau_m(k) - (\tau_{est}(k)))^2}}{\sum_{k=1}^N (\tau_m(k) - \bar{\tau}_m(k))^2} \quad (18)$$

where the number of time points per test is denoted by k . $\tau_{est}(k)$ and $\tau_m(k)$ respectively represent the estimated torque and the actual measured torque. $\bar{\tau}_m(k)$ denotes the average value of the actual measured torque.

The average *GoF* values and standard deviations (SD) of extension and flexion torques for each subject were $85.74 \pm 5.73\%$ and $90.86 \pm 2.3\%$, respectively. This is acceptable for the experiment, especially considering modeling uncertainty in the musculoskeletal model and the effects of inertia and damping of the torque sensor.

2.3 Control schemes

In the joint coordinate system, the impedance controller adjusts the output state of the robot according to the external force signal applied by the operator (Tsumugiwa 2004), which is calculated by



Fig. 5 Musculoskeletal model validation experiment

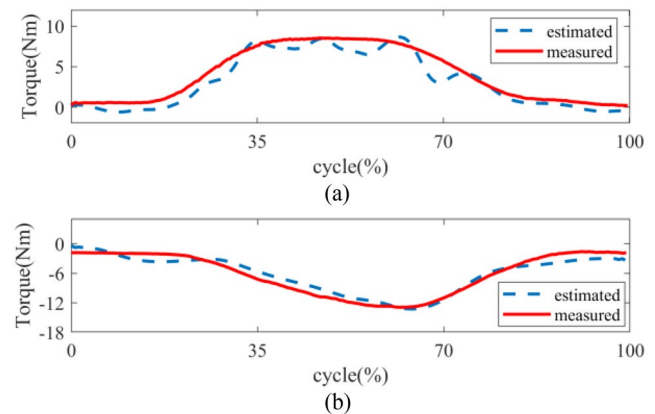


Fig. 6 Comparison of musculoskeletal model estimates and measured torques during shoulder extension and flexion: (a) extension and (b) flexion

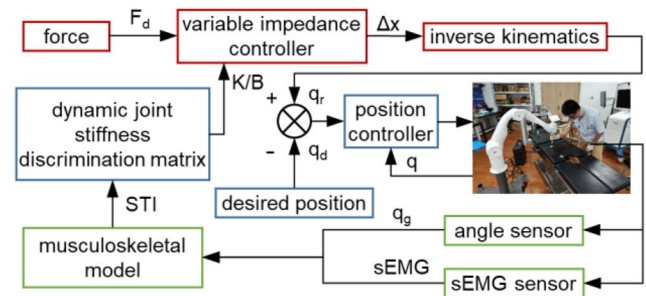


Fig. 7 Human-robot collaboration control framework

$$M_d \ddot{\Delta x} + K_d \cdot \Delta x + B_d \cdot \dot{\Delta x} = F_e \quad (19)$$

where $K_d, B_d \in R^{6 \times 6}$ are the desired stiffness and damping of the robot, which are obtained based on the stiffness of the human arm in Eq. (16) and Eq. (17). $M_d \in R^6$ is the environmental inertia parameter. $F_e \in R^6$ and $\dot{\Delta x} \in R^6$ are the external force and the velocity variation of the robot respectively.

Moreover, the $\Delta \dot{x}$, Δx can be defined by

$$\Delta \dot{x} = \dot{x}_d - \dot{x}_r, \quad \Delta x = x_d - x_r \quad (20)$$

where the x_d is the desire position and the x_r is the reference position, which can be calculated through the desired dynamics of the robot manipulator.

Lyapunov function is defined as

$$V = \frac{1}{2} M_d \dot{\Delta x}^2 + \frac{1}{2} K_e \Delta x^2 \quad (21)$$

Where K_e is the environmental stiffness, which is a positive number, and M_d is the environmental inertia parameter. Taking the first-order derivative of V , it is obtained that

$$\dot{V} = M_d \dot{\Delta x} \ddot{\Delta x} + K_e \Delta x \dot{\Delta x} \quad (22)$$

According to Eq. (19), it can be obtained that: $M \ddot{\Delta x} = F_e - K_d \Delta x - B_d \dot{\Delta x}$, and substituting it into Eq. (21)

$$\dot{V} = \dot{\Delta x} (F_e - K_d \Delta x - B_d \dot{\Delta x}) + K_e \Delta x \dot{\Delta x} \quad (23)$$

Since $F_e = -K_e \Delta x$, substituting it into the Eq.(22) to eliminate F_e , it is obtained that

$$\dot{V} = -\dot{\Delta x} (K_d \Delta x + B_d \dot{\Delta x}) \leq 0 \quad (24)$$

It can be seen that since K_d and B_d are both positive definite, their first-order derivatives are evidently non-positive. According to the stability criterion, it can be proven that the adaptive impedance control method proposed in this study is stable.

The incremental position of the robot is calculated by referring to the joint velocity, further to realize the motion control of the robot, the calculation process is given by

$$q[kT] = q[kT - T] + J^\dagger V_{ref} \Delta T \quad (25)$$

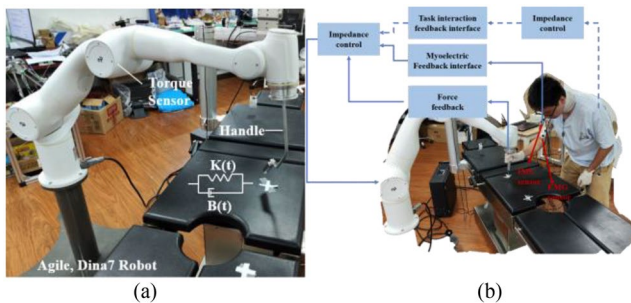


Fig. 8 Experimental setup: (a) experimental platform and (b) physical human-robot interaction strategy

where $J^\dagger$ is the Jacobian matrix generalized inverse and ΔT is the admittance control loop period, V_{ref} is the reference velocity of the robot end-effector.

3 Experiment

3.1 Experimental setup

To verify the performance of this control method in open space, we design experiments divided into two parts: free trajectory and tracking trajectory of the robot manipulator. The experimental setup is described in Fig. 8(a), where a 7-DOF robot (Diana7, Agile Robots) is used. The robot is controlled by a control cabinet with a sampling frequency of 75 Hz. Figure 8(b) illustrates the physical human-robot interaction strategy in the experiment. The sEMG signals of the human body are collected using an EMG acquisition system (Noraxon TeleMyoSystem-1400 A) with pre-gelled Ag/AgCl electrode pads at a sampling frequency of 1000 Hz, while the arm posture is recorded using an IMU sensor (Xsens Motion Capture System).

In this study, an open-space tracking task is performed to validate the proposed algorithm. The proposed human robot interaction strategy is evaluated based on average force, task execution time, trajectory error and energy consumption. The impedance parameters under different control modes, determined through preliminary experiments are: high-impedance control stiffness is selected as 100 N/m, and the damping is calculated as 14Ns/m. The low-impedance control stiffness is set to 10 N/m, and the damping is chosen as 4.43Ns/m. The variable impedance control mode calculates the stiffness and damping values in real-time based on the human sEMG signals.

3.2 Experimental task

To verify the joint variable impedance control, we design the following experiments in Fig. 9. The experiments include free trajectory and tracking trajectory in open space. The operator drags the robot out from the set origin through two reference paths and then drags the robot back to the origin. The two reference paths are rectangular and triangular. The origin and reference paths are set along the Y-axis direction of the robot. Two participants (adult males, age 25) repeated the task six times each under low impedance, high impedance, and variable impedance control methods, resulting in a total of 12 sets of experiments. The information such as execution time, end force, end velocity, sEMG, and position is recorded.

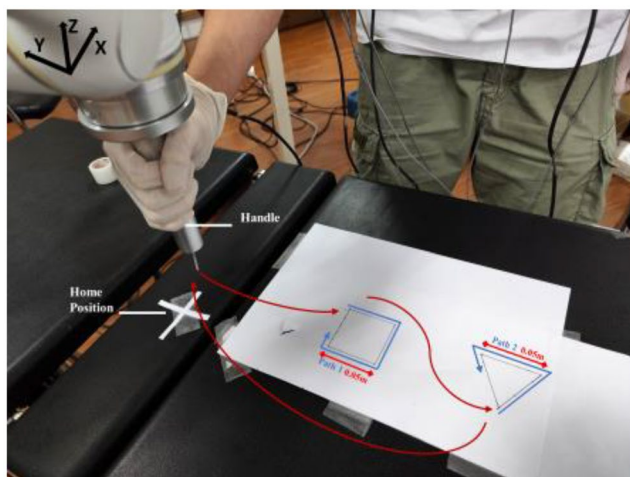


Fig. 9 Experimental reference path setting

4 Results

To evaluate the joint variable impedance control based on sEMG estimation, we design robotic manipulator dragging experiments in open space and compared the average force, task execution time, trajectory error, and energy consumption under high-impedance control, low-impedance and variable-impedance control.

The results of dragging under three impedance conditions are shown in Figs. 10 and 11. The results indicate that the end maximum force and velocity are 42.85 N and 0.343 m/s for high-impedance and 39.53 N and 0.468 m/s

for low-impedance, respectively, while the end maximum force of 33.93 N is slightly below for joint variable impedance than for low-impedance, and the velocity is 0.415 m/s, which is much higher than in the high-impedance mode. For the force component, the maximum force of 41.79 N for high-impedance is higher than 31.87 N for low-impedance and 29.57 N for variable-impedance. The maximum speed of 0.343 m/s for high-impedance is lower than 0.467 m/s for low-impedance and 0.415 m/s for variable-impedance. The stiffness value under variable-impedance control is at a higher level at tracking motion, which can reduce the tracking error. It is at a lower level at the free motion, and a higher drag speed is obtained.

The changes of sEMG signal amplitude and energy consumption in the dragging process are shown in Fig. 12. Figure 12 illustrates the amplitude changes in the sEMG signal during the dragging process. When returning to the starting point at the end of dragging, the amplitude of the sEMG signal is higher compared to the beginning, indicating increased muscle activity by the operator and higher energy consumption. Under low-impedance control, the sEMG signal typically remains lower than high-impedance control because in high-impedance mode, the operator needs to elevate muscle activation and exert more force to overcome the robot's repulsion. In addition, the sEMG signal in low-impedance mode also shows points of higher amplitude because the operator needs to overcome the inertia of the robot.

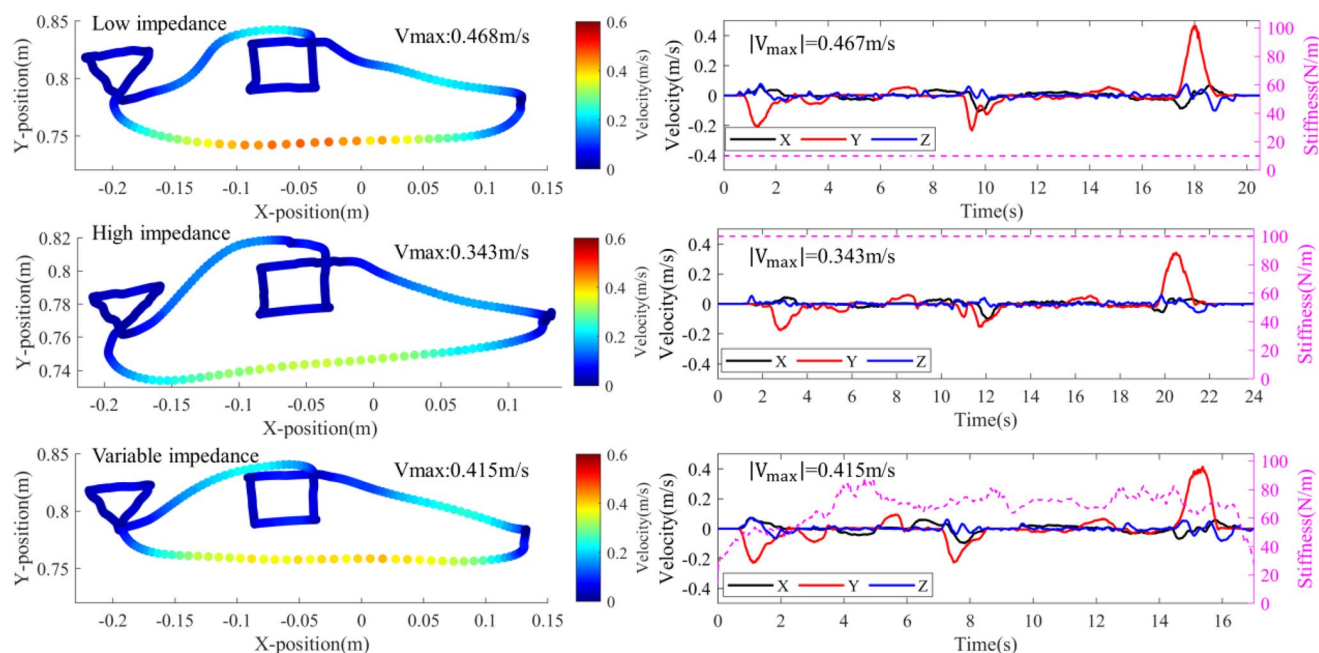


Fig. 10 End velocity and the stiffness variation in a single experiment during the dragging loop under low-impedance, high-impedance, and variable-impedance control from top to bottom. End velocities in the trajectories are shown on the left, while the velocities in the X, Y, and Z directions are displayed on the right, with stiffness changes indicated by magenta dashed lines

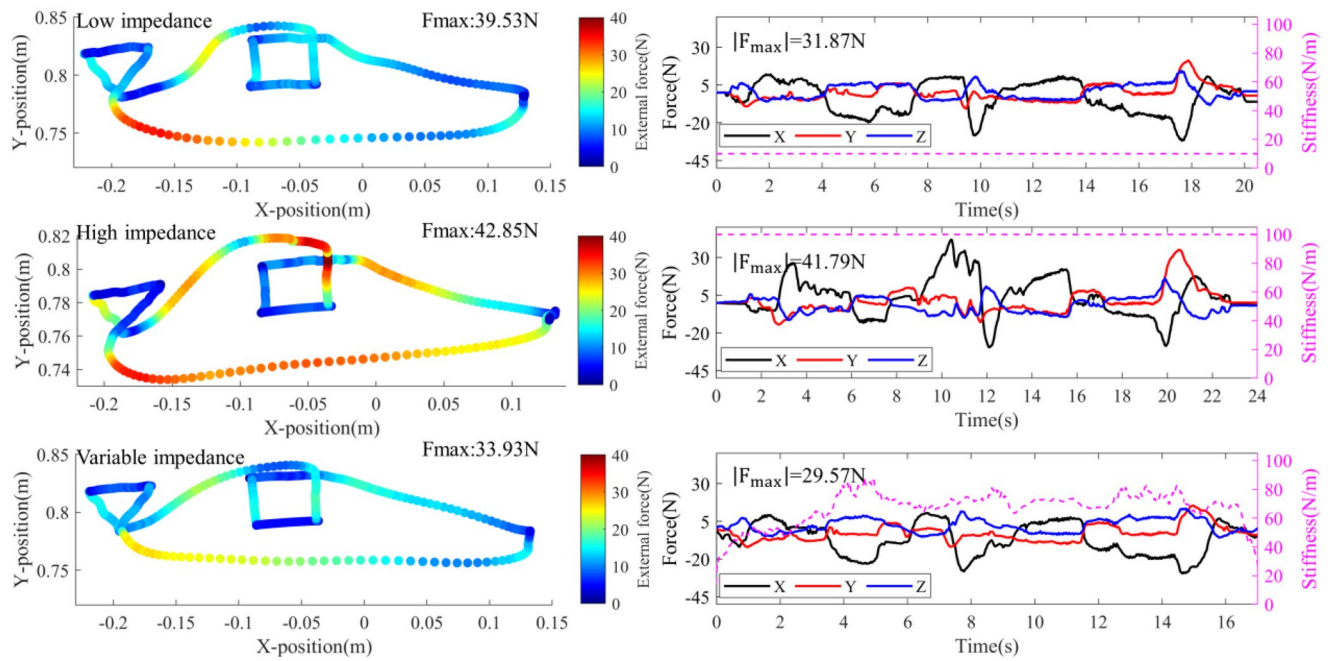


Fig. 11 End force and the stiffness variation in a single experiment during the dragging loop under low-impedance, high-impedance, and variable-impedance control from top to bottom. End forces in the trajectories are shown on the left, while the forces in the X, Y, and Z directions are displayed on the right, with stiffness changes indicated by magenta dashed lines

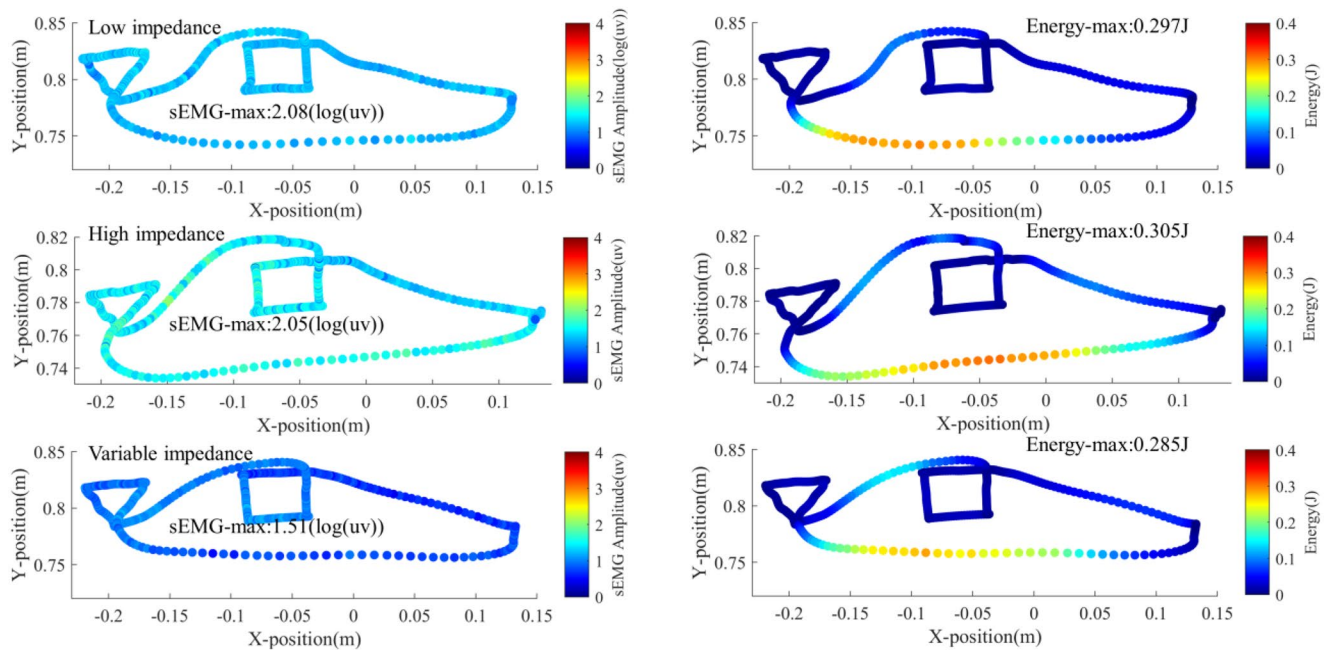


Fig. 12 sEMG amplitude and energy consumption variation in a single experiment during dragging loop under low impedance, high impedance and variable impedance control from top to bottom. The variation of the sEMG amplitude in the drag loop is shown on the left, while the variation in energy consumption is displayed on the right

During the dragging process, human energy consumption is calculated from the force and position of the robot manipulator, as shown in Fig. 12. The results indicate that in free motion, the energy consumption is low at the beginning, but it increases towards the end phase as the operator applies

more force to move the manipulator faster. Additionally, the results also show that the energy consumption of the high-impedance control is higher than that of the variable-impedance and low-impedance control, which is the

deficiency of the high-impedance control mode in human-robot interaction.

The above are the results from a single experiment, which may exist some randomness. Therefore, we have averaged the results of all experiments to observe the underlying variability, showing the average variation in velocity and force as well as the range of fluctuations in Fig. 13. Compared to the high-impedance mode, the variable-impedance mode exhibits lower average force and higher velocity, which are closer to the average force and velocity in the low-impedance mode. This is because the dragging of the robot manipulator in the high-impedance mode requires higher force. However, the fluctuation range of the force under high-impedance is higher than the other two, which may be caused by the muscle fatigue signal in low-impedance mode also shows points of higher amplitude because the operator needs to overcome the inertia of the robot.

The statistical results of the experiment are shown in Fig. 14. The task execution time is calculated based on the tracking motion time. In addition, the trajectory error is the difference between the total length of the reference trajectory and the total length of the tracking trajectory. The task execution time for high-impedance was 19.83 s, with an average force of 15.07 N. The task execution time is 14.58 s for variable-impedance and 15.04 s for low-impedance, and the average force is 9.68 N and 11.24 N for both low and variable-impedance. In the tracking motion, low-impedance exhibits a high error of 1.63 mm on the rectangular path. In contrast, the high and variable-impedances exhibit low errors of 1.21 mm and 0.91 mm, respectively, on

the rectangular path. However, the low-impedance shows an error of 1.16 mm on the triangular path, which is lower than the high-impedance of 1.52 mm and higher than the variable-impedance of 0.48 mm. The energy consumption is 24.95 J for the high-impedance and 17.11 J and 18.67 J for the low and variable-impedances, respectively, which are lower than that of the high-impedance. The variance of the variable-impedance control is larger than that of the low-impedance control in terms of average force, energy consumption, and trajectory error, which may be due to the fluctuation of the human sEMG signal in the same task. All experimental data follow a normal distribution by the Shapiro-Wilk test ($p > 0.05$). A two-tailed paired t-test is applied to the data under different methods, and the results show that there are significant differences between low impedance, high impedance, and variable impedance.

5 Discussion

The joint variable impedance strategy based on the sEMG signal can well estimate the intention of joint impedance change during the operator's movement. Thus, this characteristic can be transferred to the robot, so that the robot can flexibly adjust its impedance value according to different tasks to achieve better control. Based on the above experimental results, the following conclusions can be obtained: (1) The sEMG-based variable joint impedance interaction strategy can quickly change the impedance value of the robot controller, and the robot can switch between fast

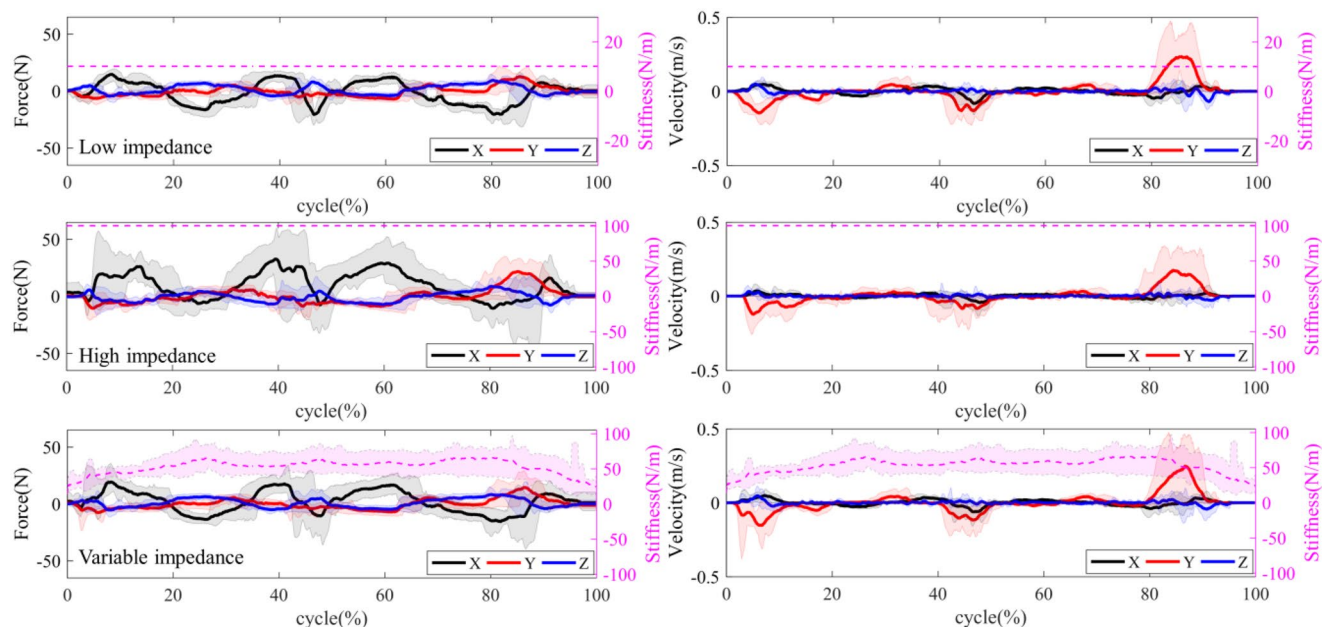


Fig. 13 The average end forces, average end speeds, and stiffness variation from 12 experiments under low impedance, high impedance, and variable impedance control. The average values of force and velocity in the X, Y, and Z directions variation by solid lines and stiffness variation by magenta dashed lines, and their fluctuation ranges across all experiments are shown by shaded region

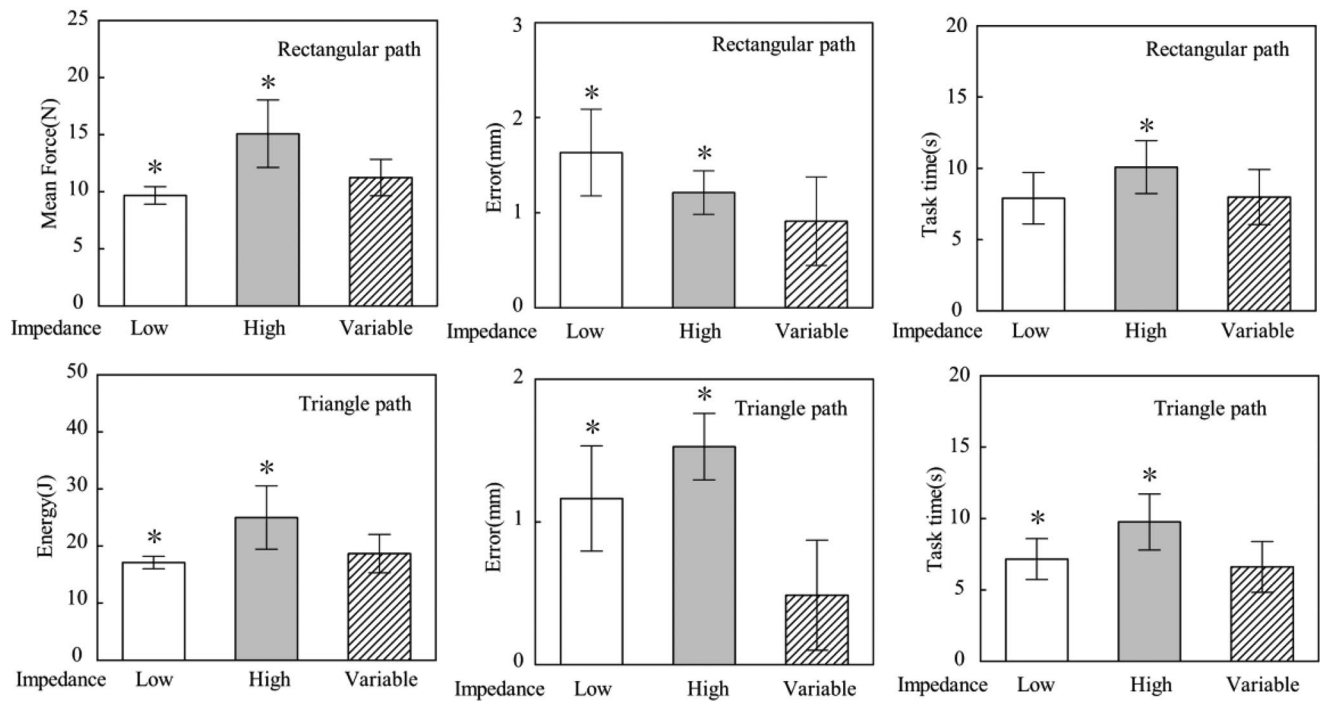


Fig. 14 The execution time, energy consumption, average force, and trajectory error under three different impedance controls (Paired test * $P < 0.05$ vs. Variable)

free motion and high precision motion according to human intention. (2) The proposed method performs as well as low-impedance control in free motion while consuming less energy compared to high-impedance control. (3) The proposed method performs as accurately as high-impedance control in the tracking motion and while excelling in repeatability and stability compared to low-impedance control.

The operator can modify the robot's joint impedance values by activating the muscle according to the actual dragging situation. When the operator wants less force and keeps dragging the robot manipulator at high speed, it will lower the arm muscle activation to a relatively low level. Therefore, at this time, the joint impedance value calculated from the sEMG signal is low, the robot exhibits good followership, and the operator can easily and quickly manipulate the robot. On the contrary, in the tracking motion, if the operator wants to track the intended trajectory better and keep the error small, the operator increases muscle activation, thus increasing the joint stiffness to improve the stability of the arm. So, at this time, the joint impedance value calculated from the sEMG signal is at a relatively high level, and the robot will exhibit excellent tracking features.

The operator needs to start and stop the robot quickly to complete the traction task as quickly as possible while maintaining adequate control performance of the robot. At this time, forces due to the limitations of its bandwidth and inertia (Hai et al. 2022). Therefore, a sudden increase in transient force can occur during motion, regardless of whether

the control method is low, high, or variable-impedance. For variable-impedance, the increase in instantaneous power is not significant and the energy consumption is smaller than at high-impedance. This is because the average impedance value of the variable-impedance is lower than the high-impedance throughout the task, resulting in a power saving effect. The high energy consumption in high-impedance mode is due to the greater damping and reactive forces generated during high-speed movements.

The sEMG signal contains rich motion information, and the damping information extracted from it can reflect the operator's operational intention. The greater the damping, the better the accuracy and stability of the motion. This can be seen from the high amplitude of sEMG concentrated in tracking motion. In tracking motion, the sEMG signal in low-impedance and variable-impedance control is usually lower than that in high-impedance control. This is because, in the high-impedance mode, the operator needs to increase muscle activation to achieve the force required to drag the robot. However, in some cases, there is a transient increase in the sEMG amplitude in the low-impedance mode compared to the sEMG amplitude in the high-impedance mode. This may be because the operator must use additional force and arm stiffness to counteract the robot's inertia. And in the triangular trajectory tracking phase, the trajectory error under low-impedance is lower than that under high-impedance. Because when the impedance value is too large, a strong reaction force is generated to hinder the dragging

process, and the robot may not be able to perform accurate trajectory tracking due to the lack of force.

In previous studies, human-robot interaction strategies mostly use variable-impedance control based on task, and experiments are done in pre-defined paths and environment (Sanchez-Magos et al. 2019). There are high and low-impedance values calculated by muscle activation and passed as output to the robot (Zhu et al. 2022). There are also continuous joint impedance values based on arm sEMG passed to the robot as output, but their experimental motion trajectories are fixed. The above methods do not apply continuously varying joint impedance values for dragging experiments in open spaces and do not reflect the compliance of physical human-robot interaction. In this study, the robots could collaborate by estimating the intention of continuously varying human joint impedance values. The method proposed in this paper has so far only been tested on a fixed trajectory. In the future, we plan to explore more application scenarios to verify its applicability in more complex tasks (Hai et al. 2022, Liu et al. 2020, Zhang et al. 2024).

6 Conclusions

With reference to the ability of humans to regulate joint impedance in various complex tasks, this paper proposes a intention-driven adaptive impedance control method for open space based on sEMG. Trajectory tracking drag experiments are conducted in open space, and the impedance transformation, speed, force, and energy performance metrics are analyzed. Experimental results demonstrate that the intention-driven adaptive impedance method balances the performance metrics between fast free motion and precise tracking motion in complex tracking motion tasks. It exhibits lower energy consumption and higher trajectory repeatability compared to fixed impedance models, enabling more natural and intuitive human-robot interaction and proving more suitable for open-space missions. Simultaneously, the adaptive impedance control strategy proposed in this study exhibits intuitive and efficient application in physical human-robot interaction tasks, showing straight forward adaptability in future industrial and domestic applications. Due to the individual variability of sEMG signals, the method proposed in this paper may not be entirely applicable to different participants. In future research, we will integrate deep learning techniques to enhance the universality of the method.

Author contributions M.H.Z. and R.Z. conceptualized the study, wrote the main manuscript text. Z.G.W. performed the investigations and H.Z.Y. curated the data, prepared figures. L.S.K. designed the methodology, and conducted formal analysis. All authors reviewed and revised the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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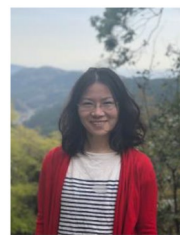
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